

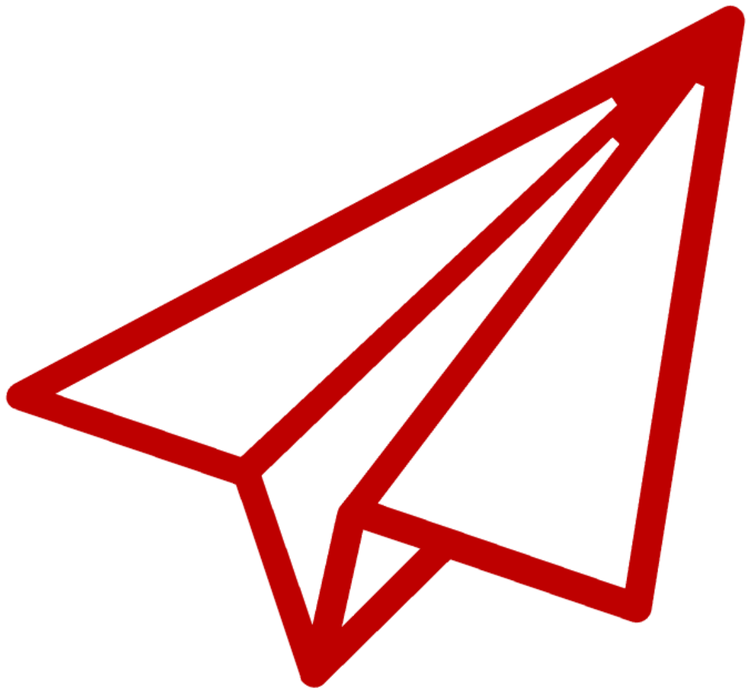
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AURIX: Long distance communication

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A.S. 2023/2024

PROJECT GOALS



To Make Two Boards Communicate Using the HC-12 Module

MOTIVATIONS AND STATE OF THE ART



Project Objective :

- Integrating the AURIX TC375 Lite Kit with the ARCELI HC-12 wireless module to have a facilitate long-distance communication.

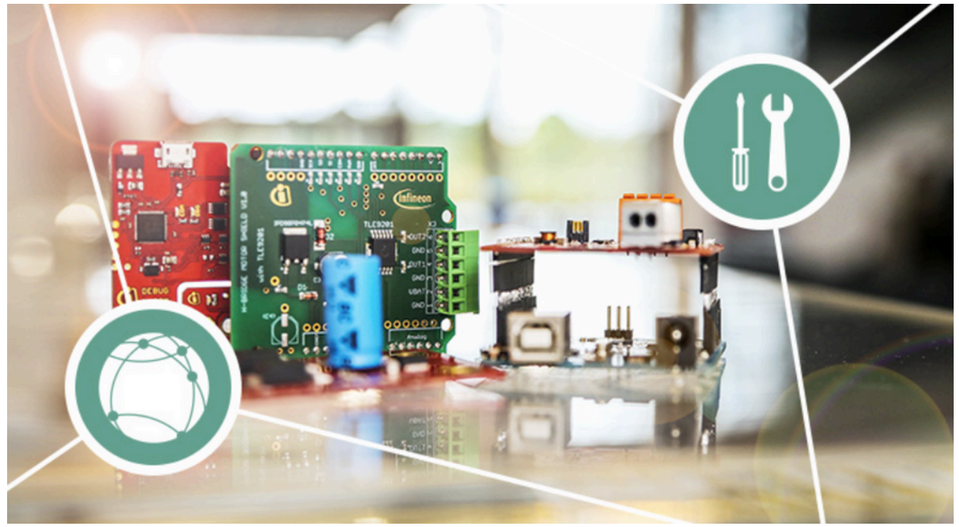
Objective Key Features :

- Utilizes the HC-12 wireless module to robust long-distance communication.
- Operates within the 433.4 MHz to 473.0 MHz frequency band, covering distances up to 1 km.
- Low power consumption, ideal for battery-operated devices.

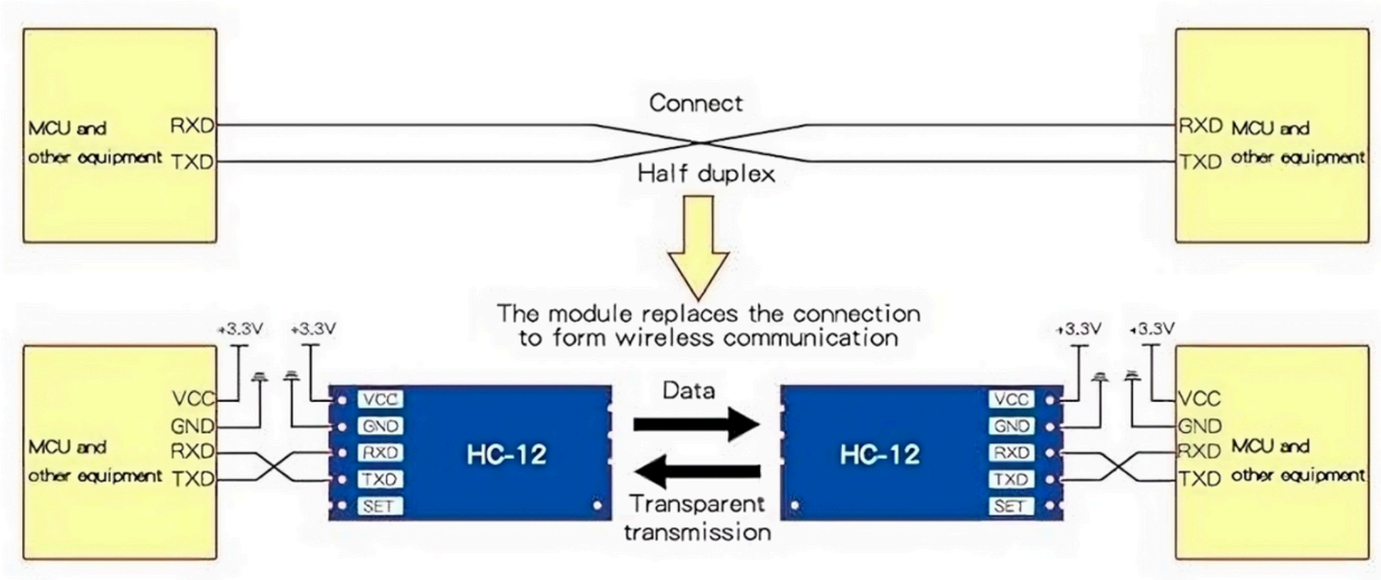
MOTIVATIONS AND STATE THE ART

Objective Advantages :

- Ability to capture real-time data and decision-making.
- Improve safety in industrial settings and support disaster emergency teams.
- Enhance efficiency in smart cities, environmental monitoring, and agriculture.
- Providing reliable communication for IoT ecosystems.



CONNECTION



HC-12 antenna module	Aurix TLC375 LiteKit
VCC	+3V3
GND	GND
RXD	P33.12
TXD	P33.13
SET	GND

ASCLIN UART 1 KIT TC375 LK project



ASCLIN_UART.C Modification:

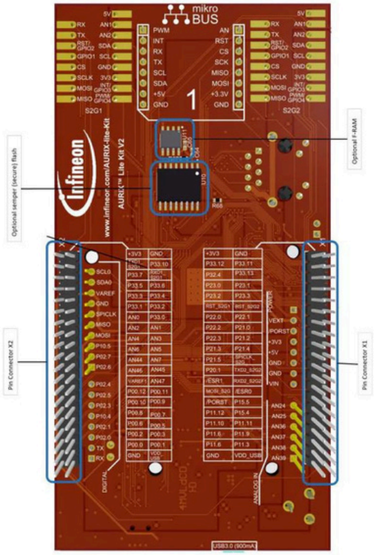
- Baud rate to 9600 bps
- Adjust the RX and TX pins
- SIZE: Size of receiving vector
- g_rx: Bytes in reception

TESTING UART COMMINICATION



Expression	Type	Value
g_rxData	unsigned char *	0x70000008, "□8□"
g_txData	unsigned char *	0x7000006c, "AT"
Add new expression		

BAUDRATE ISSUE



Baudrate ≠ 9600 bps

9600 bps



The **issue** was due to an incorrect setting of the baud rate within the baud rate function of the ASCLIN_UART_1 KIT_TC375_LK project.

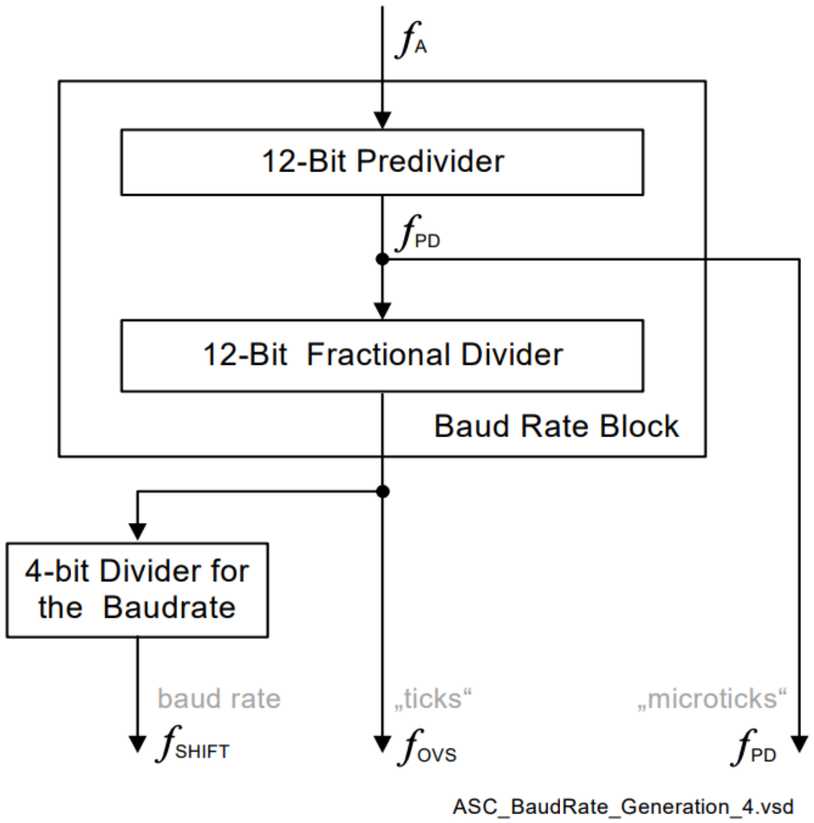
How The Baud Rate Is Calculated Inside Aurix Boards?

$$f_{PD} = f_A / (\text{BITCON.PRESCALER} + 1)$$

$$f_{OVS} = f_{PD} * \text{BRG.NUMERATOR} / \text{BRG.DENOMINATOR},$$

$$f_{SHIFT} = f_{OVS} / (\text{BITCON.OVERSAMPLING} + 1)$$

$$\text{BAUDRATE} = \frac{f_A \times \text{NUMERATOR}}{(\text{PRESCALER} + 1) \times \text{DENOMINATOR} \times (\text{OVERSAMPLING} + 1)}$$



ASC_BaudRate_Generation_4.vsd

Read and Write Function :

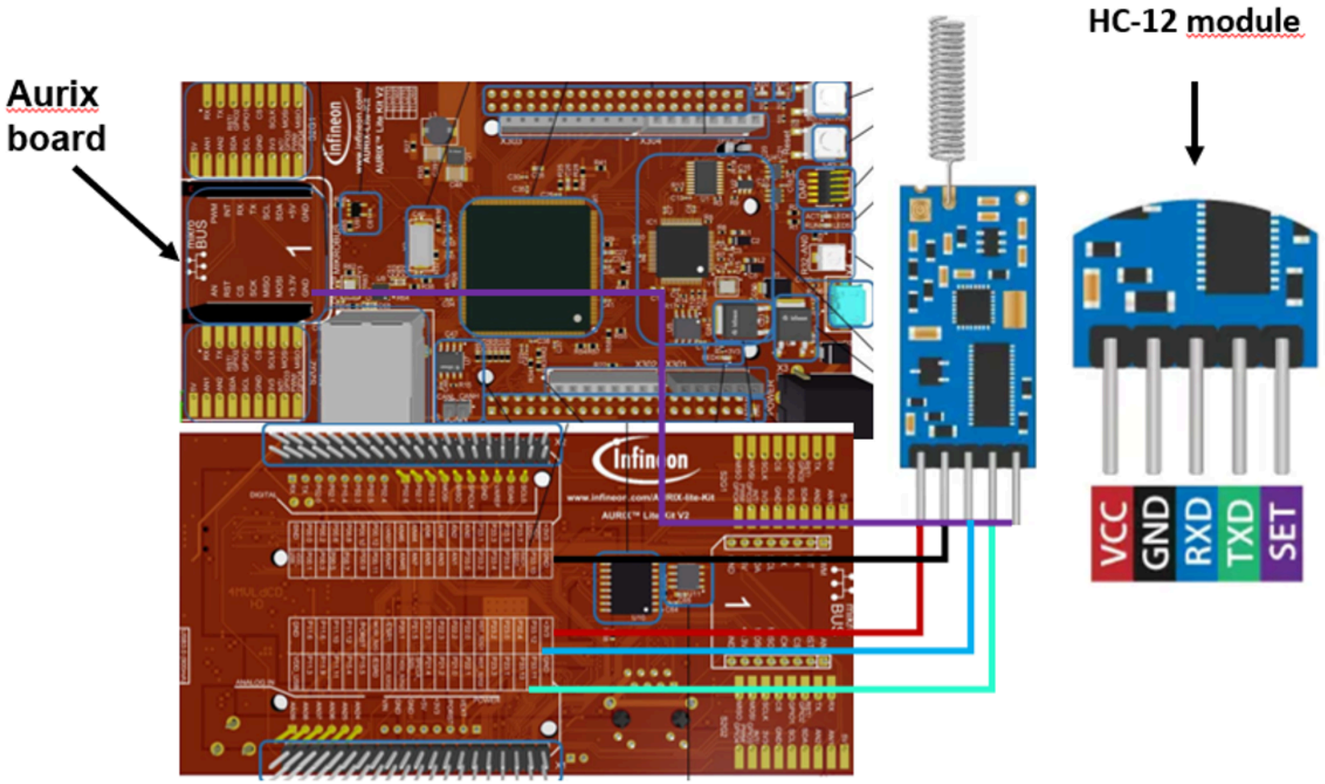
```
IfxAsclin_Asc_write(&g_ascHandle, g_txData, &g_count, TIME_INFINITE);

IfxAsclin_Asc_read(&g_ascHandle, g_rxData, &g_rx, TIME_INFINITE);
```

Sending and Receiving “Hello” Message:

Expression	Type	Value
> g_rxData	unsigned char *	0x70000004, "Hello"
Add new expression		

CONCLUSION



Possible Applications:

- To control the lights
- To turn on electronic devices
- To remotely control a vehicle
- To send temperature data



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THANK YOU FOR YOUR ATTENTION

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